Project Proposal : Track 2

Title: Iterative Verification and Feedback-Driven Repair of LLM-Generated Counterfactual Explanations

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Motivation and Research Question

The paper *Can LLMs Explain Themselves Counterfactually?* (EMNLP 2025) investigates whether large language models (LLMs) can generate counterfactual explanations that accurately reflect changes in their classification behavior. A counterfactual modifies an input to obtain a desired prediction, such as changing a negative movie review so that a sentiment classifier predicts positive. The paper highlights that generated counterfactuals may fail to achieve the intended prediction, revealing a gap between plausible explanations and actual classifier behavior.

Our Extenstion

Investigates whether verification and feedback-driven repair can improve counterfactual validity. Instead of accepting the first generated counterfactual, we will evaluate it using a fixed classifier and return the observed prediction to the LLM when the target label is not achieved. The LLM will revise its candidate until success or a predefined retry limit.

We will also investigate whether feedback accumulated from earlier failures can improve counterfactual generation for unseen inputs.

Research questions:

- **Q1:** Does iterative repair improve validity compared with one-shot generation and multi-candidate selection?
- **Q2:** Can feedback from previous failures improve counterfactual validity on unseen examples?
- **Q3:** What trade-offs arise between validity, input modification, and computational cost?

Proposed Method

Given an input, its predicted label, and a desired target label, the LLM generates a counterfactual that is evaluated by a fixed classifier. If the prediction does not match the target, the observed failure guides the next generation attempt.

We will compare three approaches:

1. **One-shot generation (baseline):** Generate and evaluate a single counterfactual without revision.
2. **Multi-candidate selection (baseline):** Generate multiple independent candidates and select a valid candidate with minimal modification, if available.
3. **Iterative verification and repair (proposed extension):** Verify each candidate and use failed predictions to guide revisions until success or the retry limit.

Additionally, we will explore **feedback memory**, which summarizes lessons from earlier failures into reusable guidelines for generating counterfactuals on new inputs. Its effectiveness will be evaluated on held-out examples, separate from those used to develop the guidelines.

Experimental Evaluation Plan

We will initially use the SST-2 sentiment classification dataset and a fixed pretrained classifier. All approaches will use the same dataset split, generation model, target-label selection procedure, and comparable generation budgets.

We will measure:

- **Validity and success rate:** Whether candidates achieve the target label and whether a valid counterfactual is found within the retry limit.
- **Edit distance:** The extent of modification relative to the original input.
- **Computational cost:** LLM calls and token usage per example.

We will also examine the effect of retry limits and compare feedback memory against iterative repair without persistent feedback. Semantic preservation will be assessed if implementation time permits.

Hypothesis and Expected Contribution

We hypothesize that iterative repair will improve counterfactual validity over one-shot generation, while multi-candidate selection may offer a competitive alternative at additional generation cost. Feedback memory may further improve validity on unseen examples.

The expected contribution is an empirical evaluation of **feedback-driven counterfactual repair and cross-example feedback transfer**, comparing their effectiveness, modification requirements, and computational costs against simpler baselines.

Reference

- Dehghanighobadi, Zahra, Asja Fischer, and Muhammad Bilal Zafar. 2025. *Can LLMs Explain Themselves Counterfactually?*. *Proceedings of EMNLP 2025*, 7787–7815. Association for Computational Linguistics.
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